

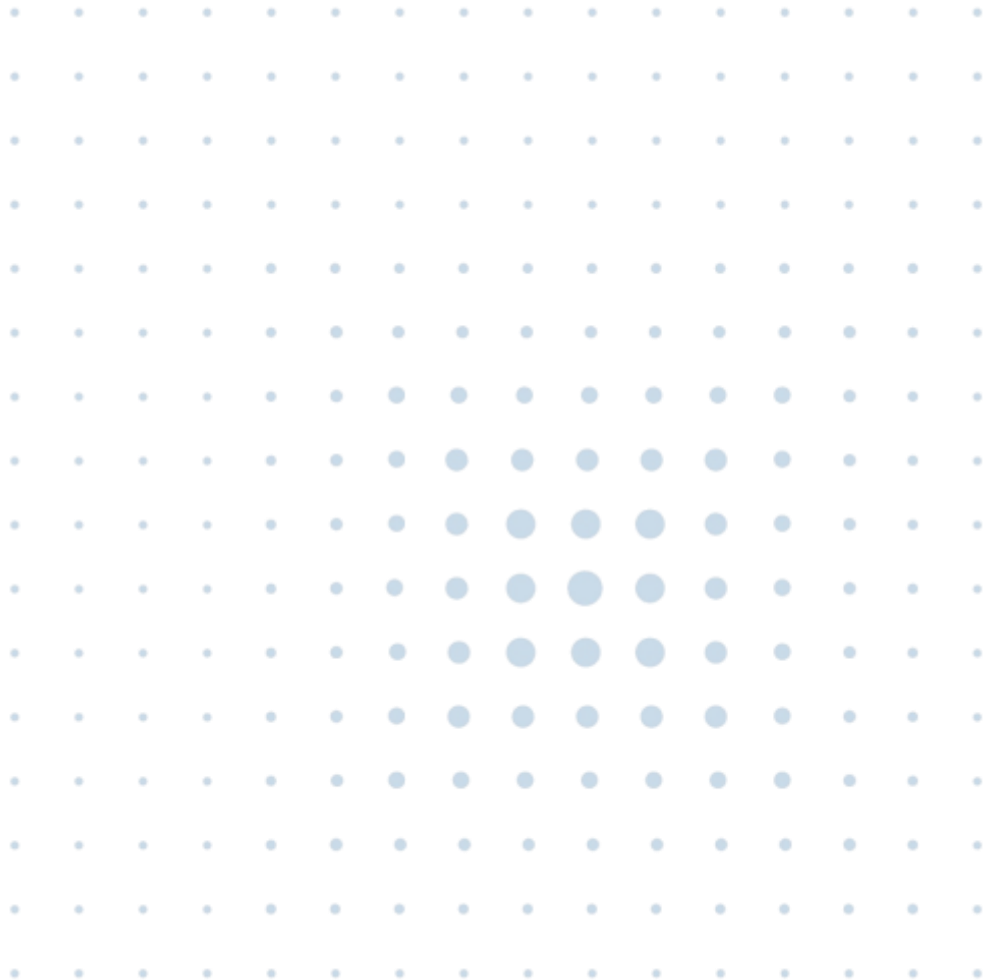
Hailo-15L SBC 2.x

Single Board Computer Kit

Quick Start Guide

Revision 1.2

March 2026



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Revision History

This section describes top-level differences in the versions of this document.

Revision	Date	Description
0.1	July 2025	First release
0.2	August 2025	Updated information on recommended camera sensor modules
0.3	October 2025	Updated descriptions and images for Hailo-15L SBC Rev 2.x Updated Section 2.3.8.2 Programming the eMMC Updated information on recommended camera sensor modules including link to Approved Vendor List. Updated the Hailo-15L SBC kit contents list
1.0	February 2026	Simplified in Section 2.3.4, the emmc_flash_program command by shortening the file paths; Added Camera Viewer in Section 3.0
1.1	February 2026	Align with Vision Processor Software Package version 1.10.1
1.2	March 2026	Align with Vision Processor Software Package version 1.11.0

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1. Overview

1.1. Introduction

This guide provides an overview of the Hailo-15L SBC 2.x, a comprehensive platform for developing and prototyping smart cameras powered by the Hailo-15L AI vision processor. Its powerful AI capabilities enable both AI-powered image enhancement and efficient, full-scale processing of multiple complex deep learning applications. For a complete description of the Hailo-15 features and capabilities, refer to the **Hailo-15 Feature-list and capabilities¹** document.

The Hailo-15L SBC showcases high-quality video streaming and advanced AI inference capabilities, including:

- Direct connection to supported camera sensors.
- Versatile interfaces for connecting, testing, and power measurement
- Easy integration with existing customers' platforms.
- Seamless integration with popular software frameworks.

The Hailo-15L SBC kit contains:

- Single Board Computer (SBC) rev. 2.x with Hailo-15L
- 12V DC power supply, microSD™ card, spacers, heatsink for the Hailo-15L, USB to USB-micro cable
- Complete Yocto Project® based Linux® distribution, a full stack DSP, libraries, Board Support Package (BSP), and drivers.

This guide is intended to help users efficiently utilize the Hailo-15L SBC.

It provides essential hardware setup and software configuration steps, as well as instructions for executing the Hailo-15L demo application.

Note:

All photos and descriptions in this guide refer to Hailo-15L SBC kit with camera module based on IMX675 sensor.

The camera's sensor and accompanying cable are not included in the Hailo-15L SBC kit and should be ordered from Hailo separately. See Section 5.1 for further information on available camera sensor modules.

The steps described in this guide were verified using Hailo-15L SBC kit, camera module with IMX675 sensor and Intel x86-64 based PC with Ubuntu 22.04 LTS, Python 3.8.10 and 3.10.12 and pypiflash 0.6.3

¹ Available in the Hailo Developer Zone ([Link](#))

2. Hardware and Software Setup

This section details the hardware and software configuration steps required for use with Hailo-15L SBC Revision 2.x

2.1. Prerequisites

The following components are required in addition to the Hailo-15L SBC kit²:

- Ethernet cable RJ45 CAT5 or above.
- X86-64 based computer running Ubuntu® 22.04 ("PC").

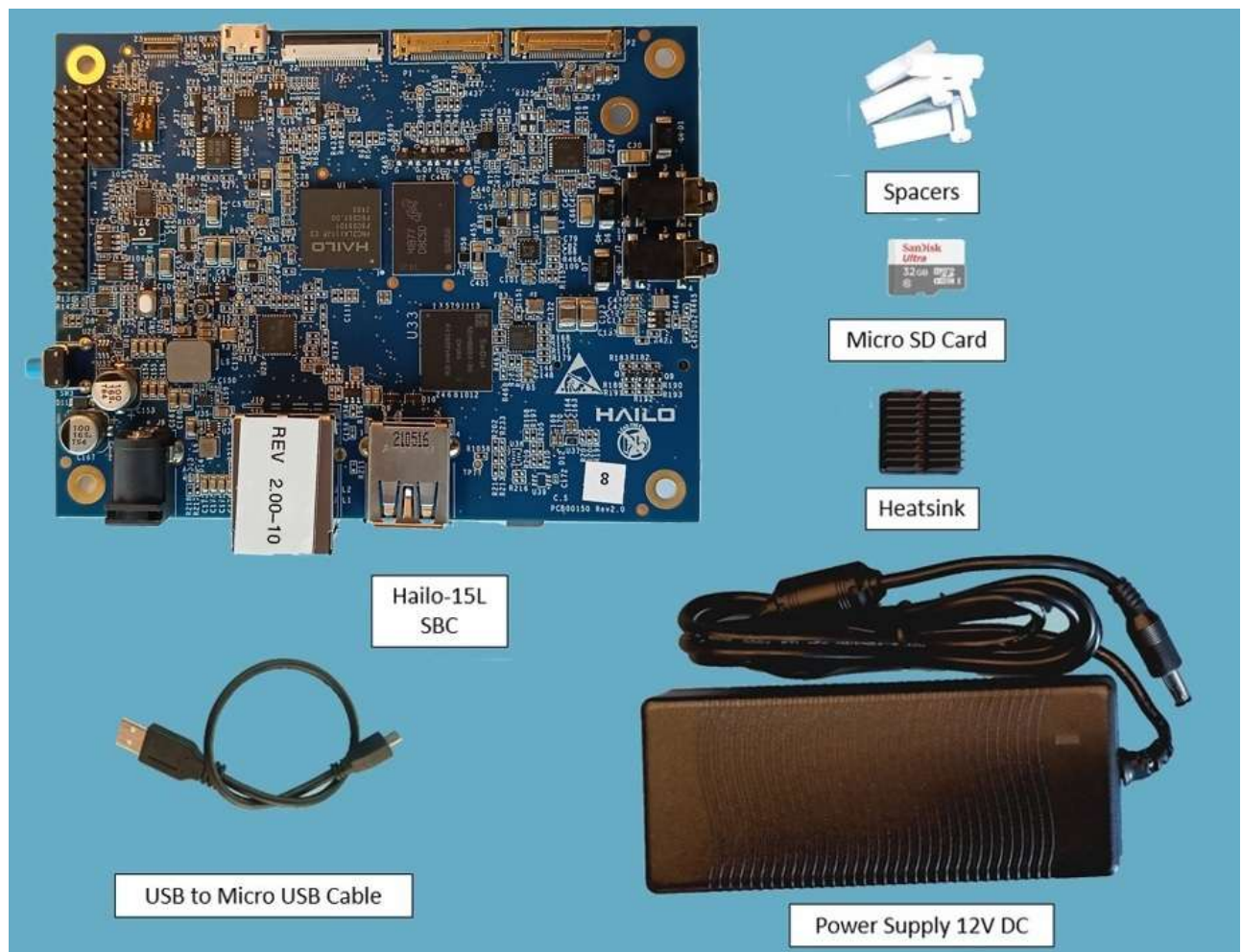


Figure 1. Hailo-15L SBC kit components

² See Table 1 for the complete list of the components included in the Hailo-15L SBC Kit

2.2. Hailo-15L SBC Key Components

Figure 2 and 3 below display the front and rear views of the Hailo 15L SBC, highlighting key component names.

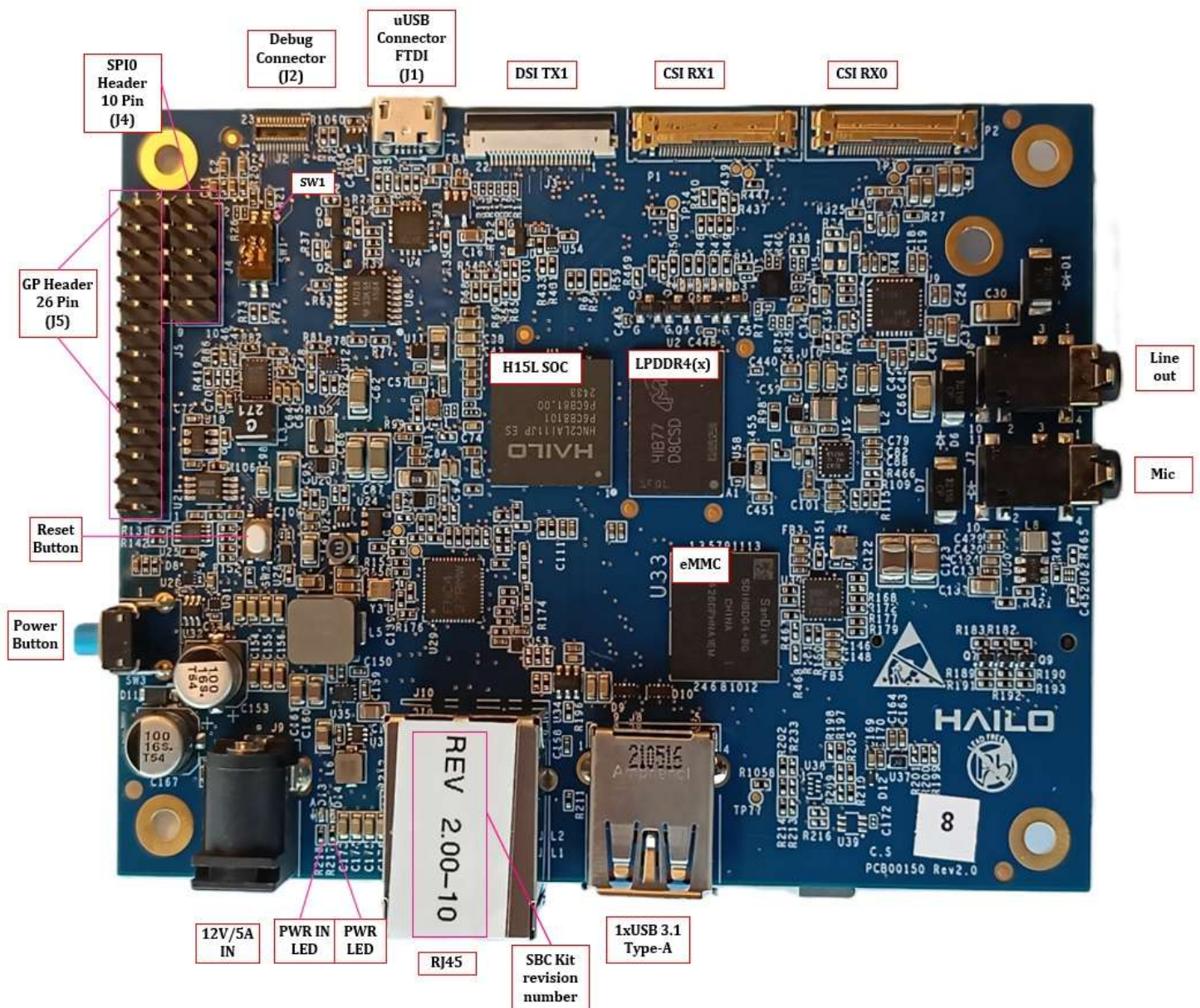


Figure 2. Hailo-15L SBC front view

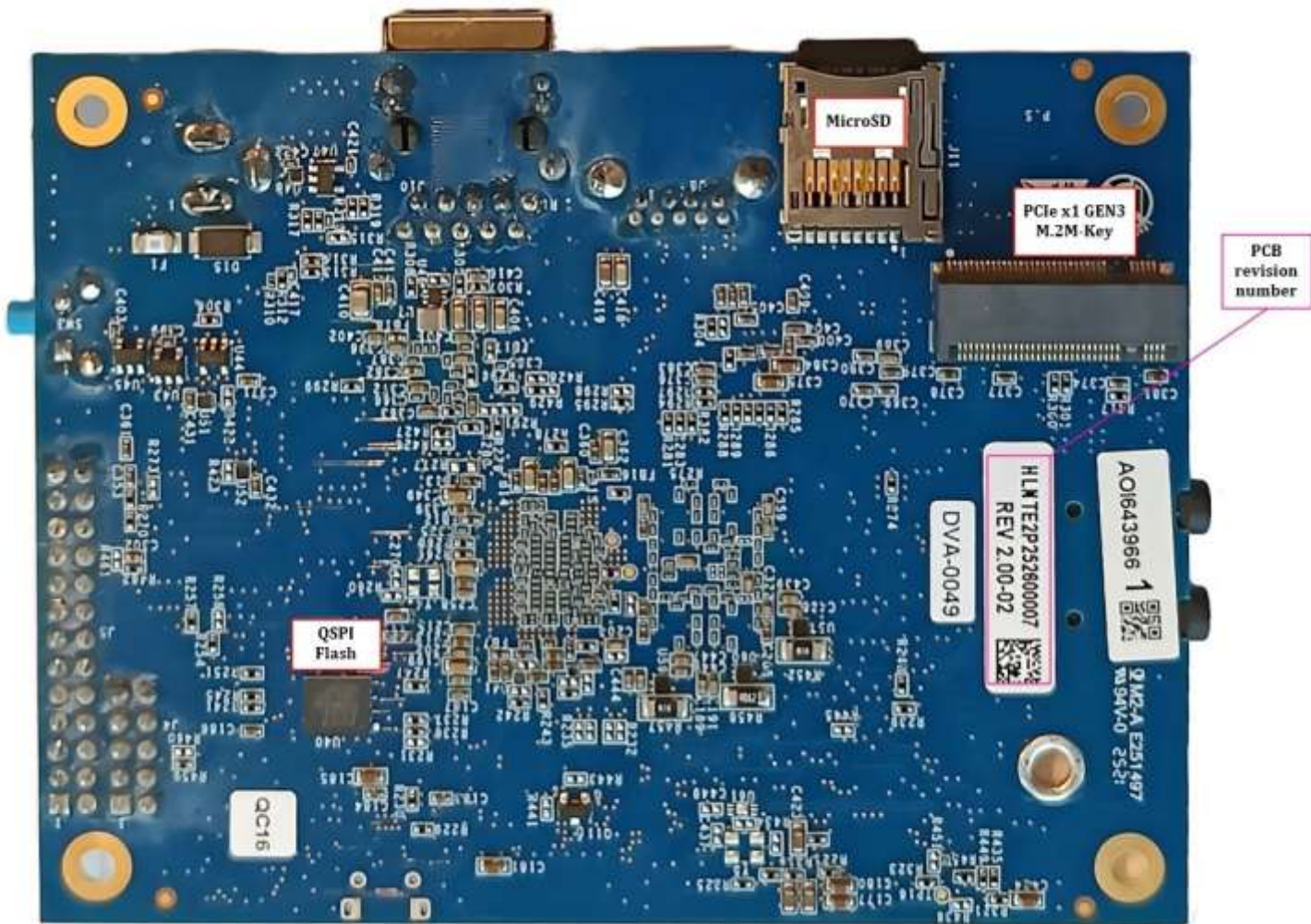


Figure 3. Hailo-15L SBC rear view

2.3. Hailo-15L SBC Software Image

The Hailo-15L SBC includes a Hailo **Vision Processor Software** image pre-loaded in its eMMC device.

Follow the steps below to verify that the Hailo-15L SBC is loaded with the latest Hailo-15L **software version**.

1. Refer to the Hailo Vision Processor Software Package Release Notes, available in Hailo Developer Zone [Link](#) to identify the most up-to-date version of the Hailo-15L software.
2. The Hailo-15L SBC is statically configured with IP address 10.0.0.1 on subnet 10.0.0.0/24. The development host must be assigned an IP address within the same subnet e.g., 10.0.0.2.
3. Login to the Hailo-15L SBC via SSH and the default static IP using the terminal command:

```
ssh root@10.0.0.1
```

(The default password is 'root')

4. Enter the command `cat /etc/os-release`

If the Hailo-15L SBC has been loaded with software version 1.11.0, continue directly to Section 2.4 Camera Sensor Module Connection, otherwise follow the steps detailed below and start in Section 2.3.1 for loading an updated software image to the Hailo-15L SBC.

2.3.1. Hailo-15L SBC Software Image Update

The Hailo-15L SBC software image update includes the following steps:

- Configure the Hailo 15L SBC to boot from UART (Section 2.3.2).
- Program the UBoot to the eMMC with the required bootloader and firmware images (Section 2.3.4). This step utilizes a recovery mechanism implemented in the Hailo-15L boot ROM.
- Reconfigure the Hailo-15L SBC to boot from the secondary program loader from eMMC (Section 2.3.6).
- Program the software image to the eMMC (Section 2.3.8).

After the update has completed, the SBC automatically reboots and runs the updated Hailo Vision Processor Software image.

⚠ WARNING ⚠

Follow the steps detailed in this guide exactly as they are described and in the correct order, failure to do so may cause damage to the hardware.

2.3.2. Hailo-15L Recovery Mechanism

The Hailo-15L has a recovery mechanism implemented in the Boot ROM. It can be used to program a blank or a corrupted eMMC device. Programming requires physical access to the Hailo-15L UART1 port.

2.3.3. Setting the Hailo-15L SBC for Boot from UART

1. Power off the Hailo-15L SBC. See Figure 4

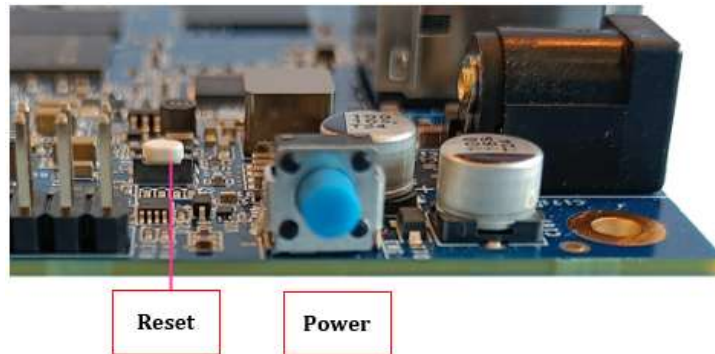


Figure 4. Power and reset buttons in the Hailo-15L SBC

2. If present using fine tweezers remove the orange tape covering the DIP switches.
 3. Using the fine tweezers, set the bootstraps to boot from UART by setting the DIP switches as shown in Figure 5).
- SW1: 1=ON, SW1: 2=OFF.



Figure 5. Setting DIP switch SW1

4. Connect the USB-to-micro-USB cable to the PC.
5. Using the buttons displayed in Figure 4, power-on the Hailo-15L SBC.
6. Using the reset button, reset the Hailo-15L SBC.
7. The Hailo-15L SBC is ready to start for the next step in the SW update.

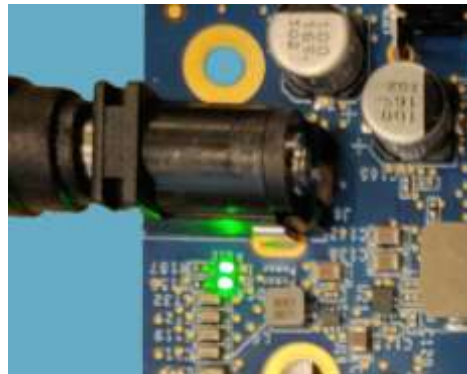


Figure 6. Adjacent green LEDs lighting up

2.3.4. Programming the U-Boot to eMMC

1. Download the Hailo-15L Vision Processor Software Package from Hailo Developer Zone (see [Link](#)) as shown in Figure 7

Package name: Hailo-15L Vision Processor Software Package

File name: `hailo_vision_processor_sw_package_1.11.0_hailo15l.tar.gz`

Note that the **Hailo-15 vision processor software package release notes** document provides briefing for the Vision Processor software package. More details for its content, structure, key components and latest features can be found in the Documents section, accessed via the Documentation tab in the Developer Zone.

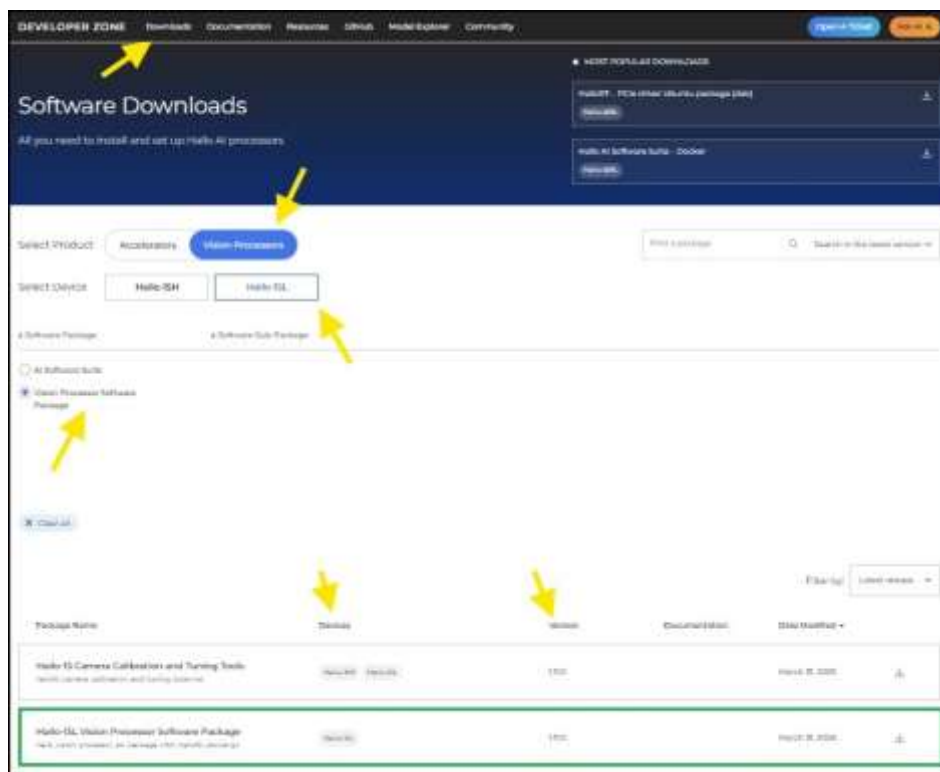


Figure 7. Software Download Page in the Hailo Developer Zone

2.3.5. Updating the Hailo-15L Images to Release 1.11.0

2. On the development host terminal, run the following commands for installation of the Hailo board tools wheel file:

```
pip install tools/hailo15_board_tools-1.11.0-py3-none-any.whl
sudo apt-get install u-boot-tools
```

Notes:

- The .whl file is part of the Hailo *Vision Processor Software Package* under the tools folder.

3. On a development host terminal

- a. Edit (or if it doesn't exist, create) the following file `/etc/udev/rules.d/99-usb.rules` and add the following line:

```
SUBSYSTEM=="usb", ATTR{idVendor}=="0403", ATTR{idProduct}=="6015", MODE="0666"
```

- b. Run the following commands:

```
sudo udevadm control --reload
sudo udevadm trigger
sudo adduser $USER plugdev
```

- c. Reboot the host machine

4. On the development host terminal, load the recovery firmware over UART using the following command:

```
uart_boot_fw_loader \
--serial-device-name /dev/ttyUSB0 \
--firmware ./prebuilt/sbc/hailo15l_uart_recovery_fw.bin --h15l
```

The .bin file is part of the *Hailo-15L Vision Processor Software Package* and is located under the *prebuilt/sbc* folder

5. On the development host terminal load the eMMC device with bootloader and firmware images using the commands below.

Note that the files mentioned in the `hailo15_emmc_program` command below are part of the Hailo *Vision Processor Software Package* and are located under the *prebuilt/sbc* folder

```
cd prebuilt/sbc

hailo15_emmc_program \
--scu-bootloader ./hailo15l_scu_bl.bin \
--scu-bootloader-config ./scu_bl_cfg_a.bin \
--scu-firmware ./hailo15l_scu_fw.bin \
--uboot-device-tree ./u-boot.dtb.signed \
--bootloader ./u-boot-spl.bin \
--bootloader-env ./u-boot-initial-env \
--customer-certificate ./customer_certificate.bin \
--uboot-tfa ./u-boot-tfa.itb \
--serial-device-name /dev/ttyUSB0 \
--uart-baud-rate 921600
```

2.3.6. Boot the U-Boot Secondary Program Loader from eMMC

1. Power off the Hailo-15L SBC.
2. Use fine tweezers to set the bootstraps to boot from eMMC by setting the DIP switches to ON as shown in Figure 8
SW1: 1=ON 2=ON



Figure 8. Setting DIP switch SW1 to boot from eMMC

3. Press the power button to power up the Hailo-15L SBC.
4. Press the reset button to reset the Hailo-15L SBC.
See Figure 9 for the location of the buttons.

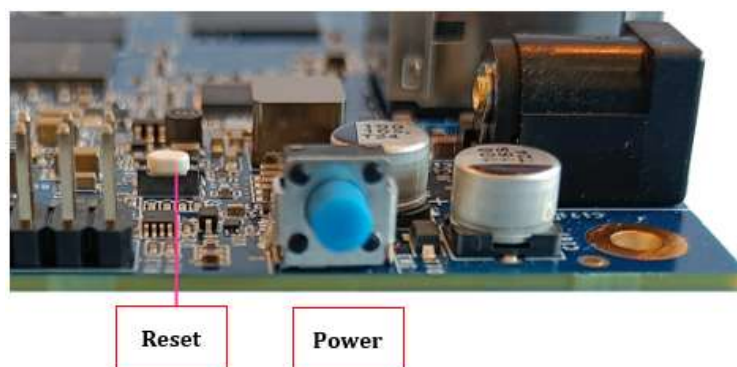


Figure 9. Hailo-15L SBC power and reset buttons

2.3.7. Ethernet Cable Connection

Connect the Ethernet cable to the development host and the Hailo-15L SBC's RJ45 connector as shown in Figure 10.

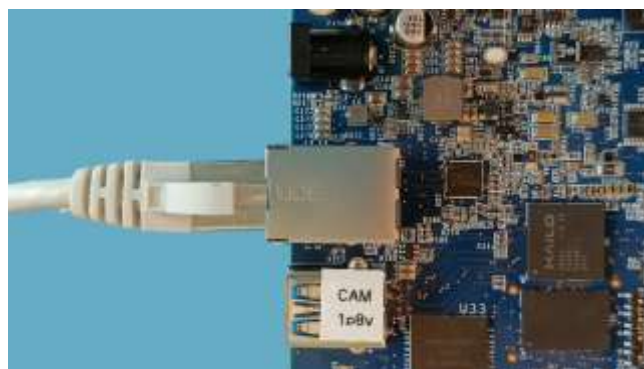


Figure 10. Ethernet RJ45 connector

2.3.8. Programming the Image to the eMMC

The following sections will describe how to program a pre-compiled image to the eMMC.

2.3.8.1. Setup TFTP Server on Host

1. Install the TFTP server required for network booting or updating:

```
sudo apt update
```

```
sudo apt install tftpd-hpa
```

2. Edit the following file `/etc/default/tftpd-hpa` and add the following lines:

```
TFTP_USERNAME="tftp"
```

```
TFTP_DIRECTORY="/var/lib/tftpboot"
```

```
TFTP_ADDRESS="0.0.0.0:69"
```

```
TFTP_OPTIONS="--secure"
```

3. Create the TFTP directory:

```
sudo mkdir -p /var/lib/tftpboot
```

```
sudo chmod -R 777 /var/lib/tftpboot
```

4. Restart the TFTP server:

```
sudo systemctl restart tftpd-hpa
```


2.3.8.2. Programming the eMMC

1. Ensure the development host Ethernet port connected to the Hailo-15L SBC, its IP address is set to 10.0.0.2 and it is on the same subnet as the Hailo-15L SBC.

(Hailo-15L Board statically defined at 10.0.0.1 with subnet 10.0.0.0/24)

2. Ensure the dip switches are set to ON. See Figure 8

3. Copy the FIT image file into the TFTP directory:

```
sudo cp fitImage /var/lib/tftpboot/
```

4. Copy the swupdate files into the TFTP directory:

```
sudo cp swupdate-image-hailo15l-sbc.ext4.gz /var/lib/tftpboot/
```

```
sudo cp hailo-update-image-hailo15l-sbc.swu /var/lib/tftpboot/
```

5. Ensure that the USB-to-Micro-USB cable between the host and the SBC is connected and open a serial console connection to the SBC by entering the command:

```
sudo picocom -b 115200 /dev/ttyUSB0
```

Note:

If a permission denied error occurs, enter the command:

```
sudo chmod 777 /dev/ttyUSB0
```

and then enters the command:

```
sudo picocom -b 115200 /dev/ttyUSB0
```

6. Reset the Hailo-15L SBC, with the reset button. See Figure 9.
7. The U-Boot menu will appear as presented in Figure 11.
Select "eMMC Board Init" from the U-Boot menu



Figure 11. U-Boot Menu

8. U-Boot will now download the SW update and will begin the update procedure.

9. Wait for the procedure to be completed by observing the traces below, followed by an automatic device reboot. See Figure 12

```
[INFO] : SWUPDATE running : [endupdate] : SWUpdate was successful !
[DEBUG] : SWUPDATE running : [postupdate] : Running Post-update command
[TRACE] : SWUPDATE running : [unlink_sockets] : unlink socket /tmp/swupdateprog
[TRACE] : SWUPDATE running : [unlink_sockets] : unlink socket /tmp/sockinstctrl
SWUpdate finished
Rebooting...
[ 137.347110] reboot: Restarting system
```

Figure 12. SW update procedure completion

2.4. Camera Sensor Module Connection

It is recommended to use the IMX675 Camera Sensor Module with the Hailo-15L SBC kit.

1. Confirm that both SW1 DIP switches on the Hailo-15L SBC are set to ON as in Figure 13.

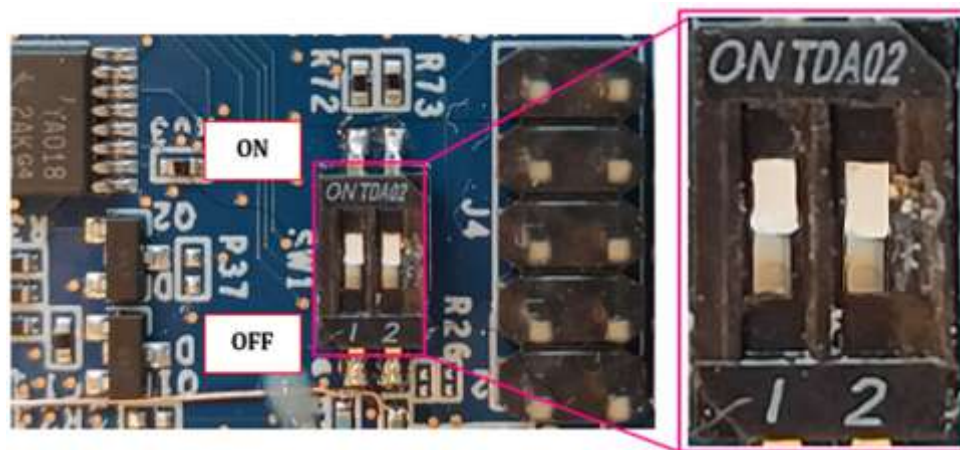


Figure 13. Hailo-15L SBC DIP SW 1

2. Connect the end of the camera sensor module cable to the IMX675 camera sensor module as shown in Figure 14.

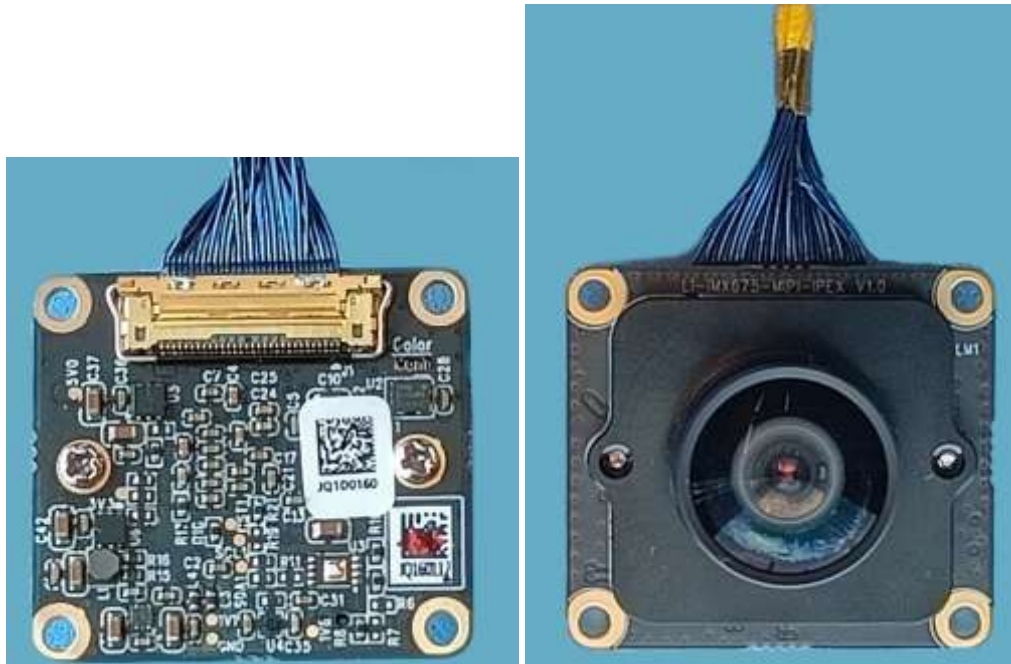


Figure 14. Sensor cable connection to the IMX675 camera sensor module

3. Connect the other end of the camera sensor module cable to the MIPI-CSI2 connector labeled 'P2' on the Hailo-15L SBC as displayed in Figure 15.



Figure 15. Camera sensor module cable connection to Hailo-15L SBC P2

2.5. Ethernet Cable Connection

If an Ethernet cable has not yet been connected, connect it to the development host and the Hailo-15L SBC's RJ45 connector as shown in Figure 16.



Figure 16. Ethernet RJ45 connector

2.6. Power Supply Connection

If power supply has not yet been connected to the Hailo-15L SBC, connect the Hailo-15L SBC-provided 12v DC power supply to the barrel jack, a single LED will light ON as shown in Figure 17

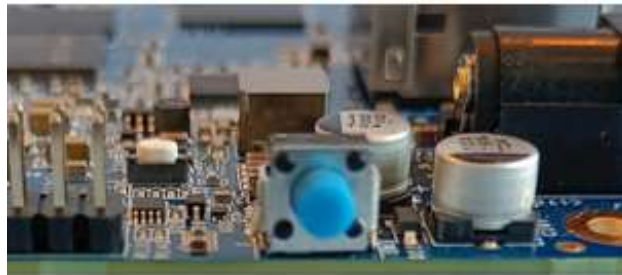


Figure 17. 12v DC power supply barrel jack

2.7. Powering up the Hailo-15L SBC

1. Press the button labeled 'PWR' to power up the Hailo-15L SBC, the adjacent green LED will light ON. See Figure 18 and Figure 19.
2. When powering on the Hailo-15L SBC (OFF → ON), a short press will turn the Hailo-15L SBC ON.

NOTE: When powering off the Hailo-15L SBC (ON → OFF), a long press will turn the Hailo-15L SBC OFF.



**Power
Button**

Figure 18. Power Button

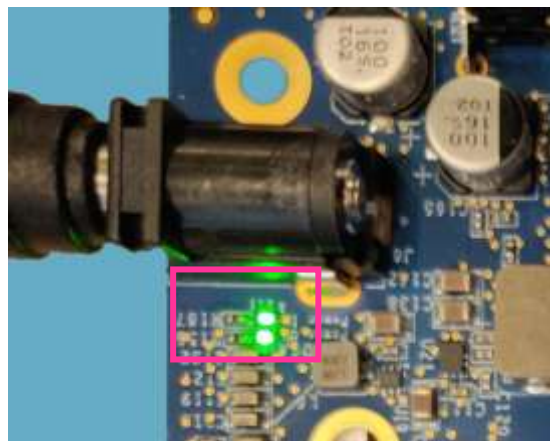


Figure 19. Adjacent green LEDs lighting up

3. Running Hailo Camera Viewer

The Hailo Camera Viewer is a video streaming interface allowing control of standard camera setup parameters such as video resolution, FPS, imaging profiles and preferences, encoding and overlay.

HDR and low-light imaging modes as well as a basic detection pipeline can be operated through functions in the AI & Image Enhancement tabs.

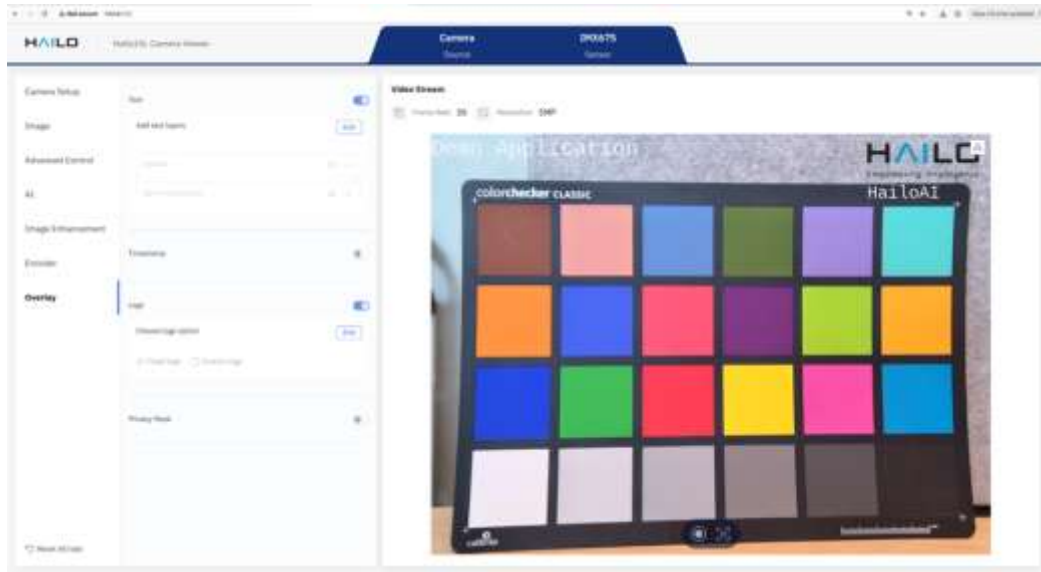


Figure 20. Camera viewer

3.1. Launch the Hailo Camera Viewer

Open a terminal on the host PC and enter the following commands:

1. Connect via SSH:

```
ssh root@10.0.0.1
```

Note: the default password: root

2. Configure the Camera Sensor

```
setup_hailo_sensor.sh
```

3. Launch the camera server:

```
camera-viewer-server
```

Leave the camera-viewer-server running; it serves the **Camera Viewer** web UI.

4. Open a browser and navigate to:

```
http://10.0.0.1/#/
```


3.2. Live Video Streaming

The Camera Viewer can be used to demonstrate full pipeline, from video capture through ISP, encoding, streaming, and display.

- In Camera Viewer, click **Play** to start the live stream.

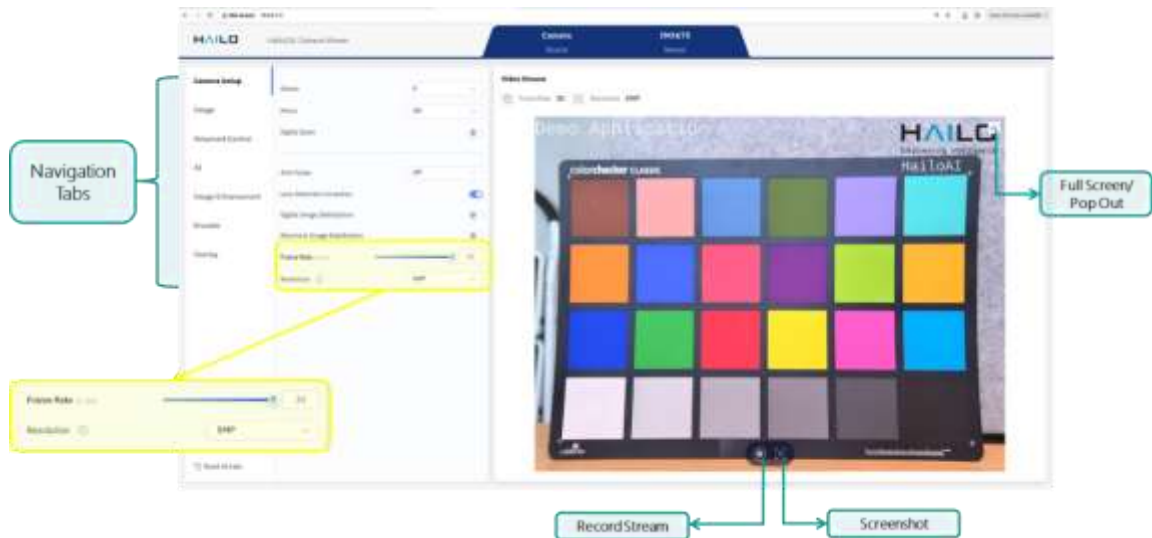


Figure 21. Camera viewer settings

- Configure:
 - Resolution e.g., 1920×1080 (FHD)
 - Frame rate e.g., 25 fps

The selected resolution and frame rate are reflected in the viewer header and the displayed stream updates accordingly. See Figure 21 and Figure 22



Figure 22. Camera viewer with updated resolution and frame rate setting

Note:

- If the stream does not start, re-run `setup_hailo_sensor.sh` and restart the camera-viewer-server.

- Basic connectivity check can be done by running
`ping 10.0.0.1`

3.3. Object Detection

Object detection can be enabled through the Camera Viewer interface.

Typical capabilities include:

- Real-time detection
- Bounding box overlay
- Class labeling



Figure 23. Camera viewer object detection setting

4. Troubleshooting

The steps described in this guide were verified using Hailo-15L SBC kit, camera module with IMX675 camera sensor module and Intel x86-64 based PC with the following software:

1. Ubuntu 22.04 LTS
2. Python 3.8.10 and 3.10.12
3. pypiflash 0.6.3

Problem	Possible Cause	Possible Solution
Hailo-15L SBC is unresponsive after power on	Bad or missing eMMC image	Make sure the DIP switches are set correctly Re-program the eMMC image
No video on the development host when the demo application is running	Video stream is blocked by the firewall	Check the configured firewall rules. This can be done by using the command: 'sudo ufw status'

5. Hailo-15L SBC Kit Contents

The Table 1 below lists the part numbers for the Hailo-15L SBC rev. 2.x Kit components.

Table 1. Hailo-15L SBC kit

Hailo Part Number	Description	Quantity
HSB2LA1C4GA	Hailo-15L SBC rev. 2.x Kit	1
	Hailo-15L SBC rev. 2.x PCB	1
MHS00025	Heatsink with Thermal-Pad-T412-28x17 20x30x15	1
	USB to micro-USB cable	1
MECH0006	Screw Round Head Philips M3 Plastic 10.00mm	4
MECH0020	Hex Spacer Threaded M3 Plastic 20.00mm	4
SDC0001	32GB Sandisk Ultra microSDHC Class10 Card 100MB/s	1
PSU000002	Desktop PSU, 100-240VAC C14 , 12V/5A 2.1x5.5x11.0mm	1

Please note that the Hailo-15L SBC kit does not include the camera module and its cable which should be ordered separately.

5.1. Supported Camera Sensor Modules

The Hailo-15L SBC supports several camera sensor modules that are fully integrated with the Hailo-15L VPU. These modules utilize a common connector, allowing for direct connection via the FAW-1233-03 cable without the need for an external adaptor board. The modules are available for order as optional accessories from Hailo.

For a complete list of all supported camera sensor modules, please refer to the Hailo-15 Approved Vendor List. This document can be found in the [Hailo Developer Zone](#)

The **IMX675 camera sensor module** is the most recommended for use with the Hailo-15L SBC

Note: The camera sensor modules purchased from Hailo include a built-in oscillator, whereas usually, the vendor's module datasheets will not include a description of it.
